0.1. Recap

- For events E, F, we wish to compute $P[E \mid F]$. The probability of the event E conditioned on (given that) F.
  - **Approach 1**: With conditioning, F can be interpreted as the *new sample space* such that for $\omega \notin F, P[\omega \mid F] = 0$.
  - **Approach 2**: $P[E|F] = \frac{P[E \cap F]}{P[F]}$.
- Multiplication rule: For events $E_1, E_2, ..., E_n$, $P[E_1 \cap E_2 \cap ... \cap E_n] = P[E_1]P[E_2|E_1]P[E_3|E_1 \cap E_2]...P[E_n|E_1 \cap E_2 \cap ... \cap E_{n-1}]$.

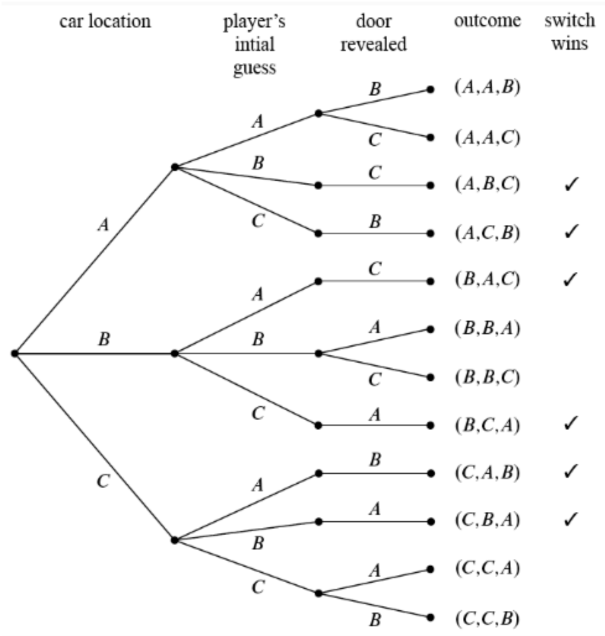
Tree diagrams

- Helpful in calculating probabilities in sequetial process (E.g. throw one dice, followed by another).
- In a tree diagram, edge-weights correspond to conditonal probabilities and leaf nodes correspond to outcomes.
- The probability of an outcome can be calculated by multiplying the relevant probabilities along a path.

1. The Monty Hall problem

Q: Suppose you're on a game show, and you're given the choice of three doors. Behind one door is a car, behind the others, goats. You pick a door, say A, and the host, who knows what's behind the doors, opens another door, say C, which has a goat. He says to you, "Do you want to pick door B?" Is it your advantage to switch your choice of doors?

- The car is equally likely to be hidden behind each of the three doors.
- The player is *equally likely* to pick each of the three doors, regardless of the car's location.
- After the player picks a door, the host must open a different door *with a goat behind it* and offer the player the choice of staying with the original door or switching
- If the host has a choice of which door to open, then he is equally likely to select each of them.



Taking a look at the given tree diagram, we can take an event E to be the event where switching doors is advantageous.

$$E = \text{Switching wins} = \{(A, B, C), (A, C, B), (B, A, C), (B, C, A), (C, A, B), (C, B, A)\}$$

$$P[(A, A)] = P[\text{Car is at A} \cap \text{Player picks A}]$$
$$= P[\text{Player picks A} \mid \text{Car is at A}] P[\text{Car is at A}] = \frac{1}{3}\frac{1}{3} = \frac{1}{9}$$
$$P[(A, A, B)] = P[\text{Door B is revealed} \cap AA] = P[\text{Door B is revealed} \mid AA] P[AA] = \frac{1}{2}\frac{1}{9} = \frac{1}{18}$$

Therefore,

$$P[E] = \Pr[(A, B, C)] + \Pr[(A, C, B)] + \Pr[(B, A, C)] +$$
$$\Pr[(B, C, A)] + \Pr[(C, A, B)] + \Pr[(C, B, A)] = \frac{1}{9} \times 6 = \frac{2}{3}$$

Q: What is the probability of winning by switching if we pick door A and door B is opened?

$$\frac{P[(C, A, B)]}{P[\{(C, A, B), (A, A, B)\}]} = \frac{\frac{1}{9}}{\frac{1}{18} + \frac{1}{9}} = \frac{2}{3}$$

This follows for any two doors.